

TARGET BRAZIL E-COMMERCE ANALYSIS



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1. Executive Summary

Target Brazil's e-commerce business demonstrated strong expansion during the analyzed period from September 2016 to October 2018. Order volume increased significantly from 329 orders in 2016 to 45,101 orders in 2017 and further to 54,011 orders in 2018, highlighting rapid customer adoption and marketplace growth.

The analysis reveals that customer demand is heavily concentrated in Brazil's economically developed southeastern states, particularly São Paulo, Rio de Janeiro, and Minas Gerais, which together contribute a substantial share of the platform's customer base and revenue generation.

Operational performance analysis identified a strong relationship between geographic distance, freight costs, and delivery times. Remote northern states such as Roraima, Paraíba, Rondônia, Acre, and Piauí experience the highest average freight costs, while southeastern states benefit from significantly lower shipping expenses and faster deliveries.

Payment analysis shows overwhelming customer preference for credit card transactions along with widespread adoption of installment-based purchasing behavior, reinforcing the importance of flexible financing options within the Brazilian e-commerce market.

Based on these findings, Target Brazil should prioritize logistics optimization in high-cost regions, continue investing in high-performing states, improve delivery estimation accuracy, and further strengthen installment payment offerings to support continued growth and customer satisfaction.

2. Business Problem Statement

Operating a nationwide e-commerce platform in a geographically large country such as Brazil creates significant challenges in customer acquisition, order fulfillment, logistics management, and payment processing.

Target Brazil must balance rapid business growth with efficient operational performance while serving customers distributed across thousands of cities and all Brazilian states. Differences in customer density, transportation infrastructure, and regional purchasing behavior make it difficult to maintain consistent service quality and profitability across the country.

The objective of this project is to use SQL and Google BigQuery to analyze customer behavior, order trends, geographic distribution, freight costs, delivery performance, and payment preferences in order to identify operational bottlenecks and business growth opportunities.

The analysis aims to answer key strategic questions:

- Is the business growing sustainably over time?
- Which regions contribute most to customer acquisition and revenue?
- How do freight costs and delivery performance vary across states?
- What payment methods do customers prefer?
- How can operational efficiency and customer satisfaction be improved?

The findings from this project are intended to support data-driven decisions in logistics planning, marketing investment, warehouse placement, and checkout optimization.

3. Dataset Overview

The analysis uses Target Brazil's public e-commerce transactional dataset consisting of multiple interconnected tables representing the complete customer purchase journey from order placement to final delivery.

The dataset covers the period from **4 September 2016 to 17 October 2018**, providing more than two years of historical order activity suitable for trend analysis, seasonality detection, and operational performance evaluation.

Tables Used

customers	Customer identifiers and location details, including city, state, and zip code prefix.
geolocation	Latitude and longitude reference data linked to zip code prefixes, used for mapping.
orders	One row per order, including status and every key timestamp from purchase to delivery.
order_items	Line level detail for each order, including price and freight_value per item.
payments	Payment method, number of installments, and payment value for each order.
order_reviews	Customer review scores and comments linked to each order.
products	Product level attributes such as category, weight, and dimensions.
sellers	Seller identifiers and location details.

The dataset includes:

- 99,000+ orders
- Customers from 2,318 cities
- Customers across all 27 Brazilian states
- Multiple payment methods
- Complete order lifecycle timestamps

This breadth of information enables both strategic and operational business analysis.

The dataset is structured as a set of related tables covering customers, orders, order items, payments, reviews, products, and sellers. Each table captures a different part of the order lifecycle, and they are connected through shared identifier columns, primarily `order_id` and `customer_id`.

4. Methodology

The project was completed entirely using SQL queries executed within Google BigQuery.

The analysis followed a structured approach consisting of five stages:

Data Understanding

Initial exploration was performed to understand table structures, column definitions, relationships, and data availability across the dataset.

Data Preparation

Multiple tables were joined using primary business keys such as `order_id`, `customer_id`, `seller_id`, and `product_id` to create analysis-ready datasets.

Exploratory Analysis

Aggregate functions, grouping operations, date functions, and conditional logic were used to identify trends in customer behavior, order growth, logistics performance, and payment preferences.

Business Interpretation

Each query output was translated into business insights rather than remaining as technical results. Particular emphasis was placed on identifying operational inefficiencies and growth opportunities.

Recommendation Development

Actionable recommendations were generated based on observed patterns to improve profitability, delivery performance, customer experience, and long-term scalability.

5. Analysis Sections

Question 1

What are the structure and characteristics of the customer dataset?

Business Objective

Understand the structure of the customer dataset and identify the type of information available for customer segmentation and geographical analysis.

SQL Query

```
SELECT
column_name,
data_type,
is_nullable
FROM `target_sql.INFORMATION_SCHEMA.COLUMNS`
WHERE table_name = 'customers'
ORDER BY ordinal_position;
```

Query Output

Row	column_name	data_type	is_nullable
1	customer_id	STRING	YES
2	customer_unique_id	STRING	YES
3	customer_zip_code_prefix	INT64	YES
4	customer_city	STRING	YES
5	customer_state	STRING	YES

Business Insight

The customers table primarily contains customer identifiers and location information such as city, state, and zip code prefixes. The dataset is well structured for performing customer distribution and regional demand analysis.

Recommendation

Maintain standardized customer location data as it plays a critical role in logistics planning, regional expansion, and targeted marketing campaigns.

Potential Business Impact

Improved customer data quality enables more accurate business decisions related to demand forecasting and supply chain optimization.

Question 2

What is the time range covered by the orders in this dataset?

Business Objective

Determine the period covered by the dataset to understand the time horizon available for trend and seasonality analysis.

SQL Query

```
SELECT
MIN(order_purchase_timestamp) AS first_order,
MAX(order_purchase_timestamp) AS last_order
FROM `target_sql.orders`;
```

Query Output

Row	first_order	last_order
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC

Business Insight

The dataset covers more than two years of business operations, providing sufficient historical data to identify growth trends, customer behavior patterns, and seasonal demand fluctuations.

Recommendation

Continue maintaining historical transaction data as longer time horizons improve forecasting accuracy and strategic planning.

Potential Business Impact

Better historical visibility leads to improved inventory management and long-term business planning.

Question 3

How many cities and states did customers order from during the period studied?

Business Objective

Measure the geographical reach of Target Brazil's e-commerce operations.

SQL Query

```
SELECT
COUNT(DISTINCT c.customer_city) AS total_cities,
COUNT(DISTINCT c.customer_state) AS total_states
FROM `target_sql.orders` o
JOIN `target_sql.customers` c
ON o.customer_id = c.customer_id
WHERE EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2018
AND EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 3;
```

Query Output

Row	total_cities	total_states
1	2318	27

Business Insight

Target Brazil successfully serves customers from thousands of cities and all major states across Brazil, demonstrating strong national presence and market penetration.

Recommendation

Identify underserved regions with lower customer penetration and explore localized marketing campaigns to increase adoption.

Potential Business Impact

Improved regional penetration can generate additional revenue while reducing dependence on a small number of states.

Question 4

Is there a growing trend in the number of orders placed over the past years?

Business Objective

Evaluate whether the business is experiencing sustainable growth over time.

SQL Query

```
SELECT  
EXTRACT(YEAR FROM order_purchase_timestamp) AS year,  
COUNT(order_id) AS no_of_orders  
FROM `target_sql.orders`  
GROUP BY year  
ORDER BY year;
```

Query Output

Row	year	no_of_orders
1	2016	329
2	2017	45101
3	2018	54011

Business Insight

Order volume increased significantly from 2016 to 2018, indicating rapid expansion of Target Brazil's e-commerce business and increasing customer adoption.

Recommendation

Continue investing in infrastructure, warehouse capacity, and logistics capabilities to support future growth.

Potential Business Impact

Proactive infrastructure investments prevent operational bottlenecks and maintain customer satisfaction during periods of rapid expansion.

Question 5

Is there monthly seasonality in the number of orders placed?

Business Objective

Identify recurring monthly demand patterns and seasonal purchasing behavior.

SQL Query

```
SELECT  
EXTRACT(YEAR FROM order_purchase_timestamp) AS year,  
EXTRACT(MONTH FROM order_purchase_timestamp) AS month,  
COUNT(order_id) AS no_of_orders FROM `target_sql.orders`  
GROUP BY year, month  
ORDER BY year, no_of_orders DESC;
```

Query Output

Row	year	month	no_of_orders
1	2016	10	324
2	2016	9	4
3	2016	12	1
4	2017	11	7544
5	2017	12	5673
6	2017	10	4631
7	2017	8	4331
8	2017	9	4285
9	2017	7	4026
10	2017	5	3700

Business Insight

The number of orders varies across months, suggesting the presence of seasonal shopping behavior and promotional periods.

Recommendation

Align inventory planning, marketing campaigns, and staffing levels with expected seasonal demand peaks.

Potential Business Impact

Improved seasonal planning reduces stockouts and minimizes excess inventory costs

Question 6

During what time of day do Brazilian customers mostly place their orders?

Business Objective

Understand customer purchasing behavior throughout the day.

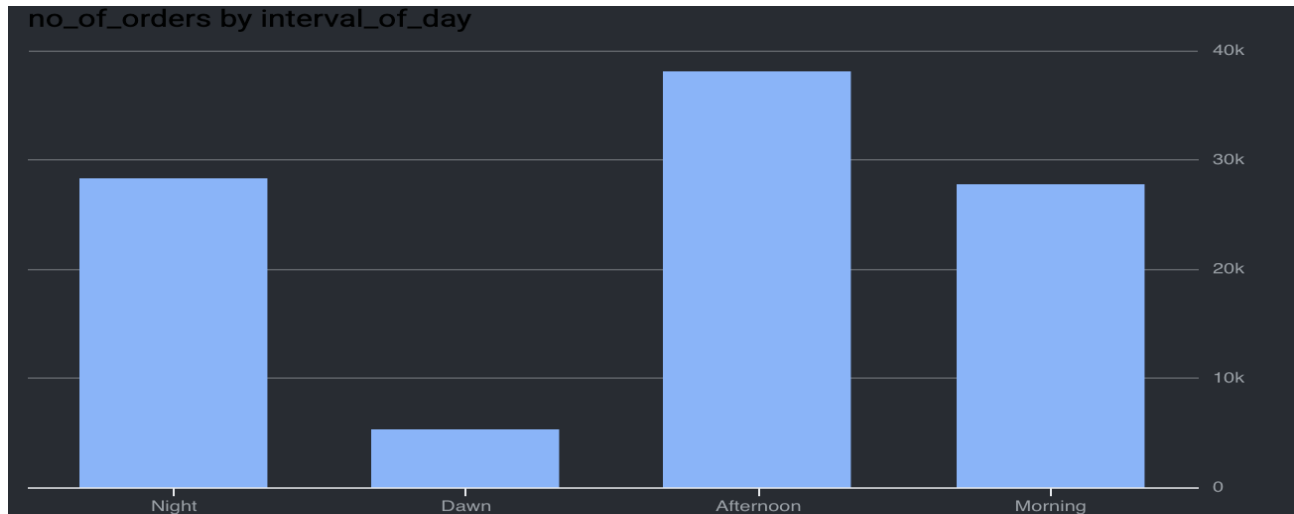
SQL Query

```
WITH cte AS (  
  SELECT  
  CASE  
    WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 0 AND 6 THEN 'Dawn'  
    WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 7 AND 12 THEN 'Morning'  
    WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 13 AND 18 THEN 'Afternoon'  
    WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 19 AND 23 THEN 'Night'  
  END AS interval_of_day  
  FROM `target_sql.orders`  
)  
SELECT interval_of_day, COUNT(*) AS no_of_orders  
FROM cte  
GROUP BY interval_of_day;
```

Query Output

Row	interval_of_day	no_of_orders
1	Night	28331
2	Dawn	5242
3	Afternoon	38135
4	Morning	27733

Visualization



Business Insight

Most customers place orders during afternoon and evening hours, while order activity remains relatively low during dawn hours.

Recommendation

Schedule marketing campaigns, advertisements, and promotional notifications during high activity periods to maximize engagement.

Potential Business Impact

Better campaign timing can improve conversion rates and customer engagement.

Question 7

What is the month on month number of orders placed in each state?

Business Objective

Analyze regional demand patterns and identify high-growth markets.

SQL Query

```
WITH cte AS
  (SELECT o.order_id, c.customer_state, EXTRACT(MONTH FROM o.order_purchase_timestamp)
  AS month,
  EXTRACT(YEAR FROM o.order_purchase_timestamp) AS year
  FROM `target_sql.orders` AS o
  INNER JOIN `target_sql.customers` c
  ON o.customer_id = c.customer_id)

SELECT customer_state, month, year, COUNT(order_id) AS no_of_orders
```

```
FROM cte
GROUP BY customer_state, month, year
ORDER BY customer_state, month, year;
```

Query Output

Row	customer_state	month	year	no_of_orders
1	AC	1	2017	2
2	AC	1	2018	6
3	AC	2	2017	3
4	AC	2	2018	3
5	AC	3	2017	2
6	AC	3	2018	2
7	AC	4	2017	5
8	AC	4	2018	4
9	AC	5	2017	8
10	AC	5	2018	2

Business Insight

Order volumes vary considerably across states, with economically stronger states contributing a larger share of overall demand.

Recommendation

Develop state-specific growth strategies instead of relying on a uniform national approach.

Potential Business Impact

Targeted investments improve marketing efficiency and maximize return on investment.

Question 8

How are customers distributed across all the states?

Business Objective

This shows where the existing customer base is concentrated, which is essential for planning warehouse locations and regional marketing budgets.

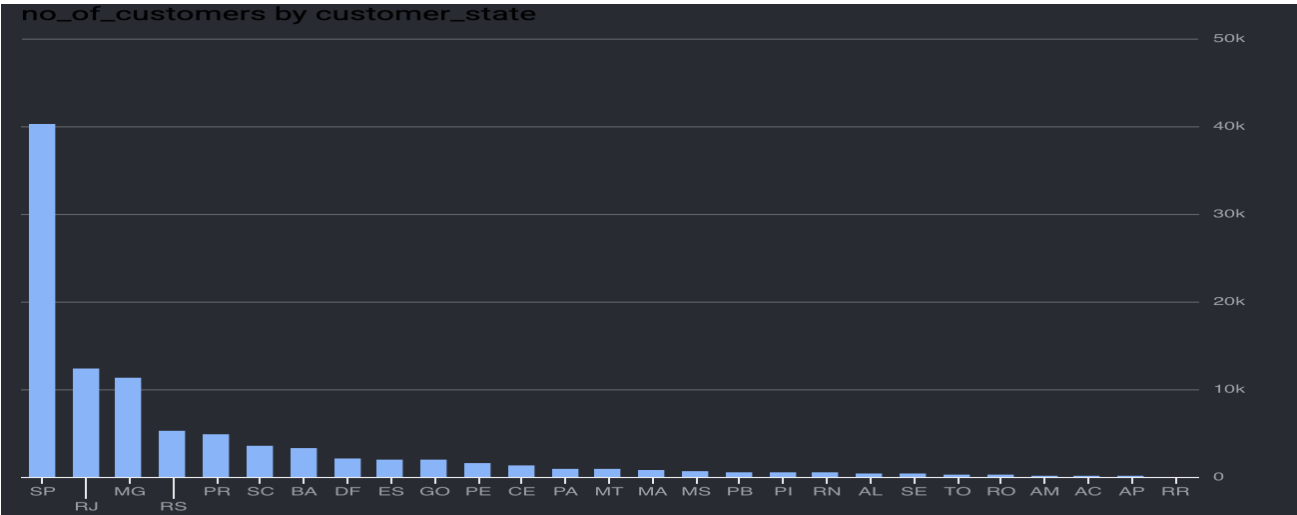
SQL Query

```
SELECT customer_state, COUNT(DISTINCT customer_unique_id) AS no_of_customers
FROM `target_sql.customers`
GROUP BY customer_state
ORDER BY no_of_customers DESC;
```

Query Output

Row	customer_state	no_of_customers
1	SP	40302
2	RJ	12384
3	MG	11259
4	RS	5277
5	PR	4882
6	SC	3534
7	BA	3277
8	DF	2075
9	ES	1964
10	GO	1952

Visualization



Business Insight

A significant proportion of customers are concentrated in southeastern states, particularly São Paulo and neighboring regions.

Recommendation

Continue strengthening operations in high-performing states while expanding customer acquisition efforts in emerging markets.

Potential Business Impact

Balanced geographic growth reduces business concentration risk.

Question 9

What was the percentage increase in order value from 2017 to 2018?

Business Objective

This measures revenue growth in monetary terms, not just order count, giving a clearer picture of the business's financial trajectory. The comparison is limited to January through August of each year so that both periods are measured on equal footing.

SQL Query

```
WITH yearly_totals AS
(SELECT EXTRACT(YEAR FROM o.order_purchase_timestamp) AS year,
ROUND(SUM(p.payment_value),2) AS total_payment
FROM `target_sql.payments` p
JOIN `target_sql.orders` o
ON p.order_id = o.order_id
WHERE EXTRACT(YEAR FROM o.order_purchase_timestamp) IN (2017,2018)
AND EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8
GROUP BY year)

SELECT
ROUND(100 *
(MAX(CASE WHEN year = 2018 THEN total_payment END) -
MAX(CASE WHEN year = 2017 THEN total_payment END)) /
MAX(CASE WHEN year = 2017 THEN total_payment END), 2) AS percent_increase
FROM yearly_totals;
```

Query Output

Row	percent_increase
1	136.98

Business Insight

Target Brazil experienced a substantial increase in order value between 2017 and 2018, indicating strong commercial performance and growing customer spending.

Recommendation

Investigate the drivers behind this growth to replicate successful strategies in future periods.

Potential Business Impact

Understanding growth drivers enables more efficient allocation of marketing and operational resources.

Question 10

What is the total and average order price and freight value for each state?

Business Objective

This identifies which states generate the most order value in absolute terms and which states have customers who tend to place higher value orders on average.

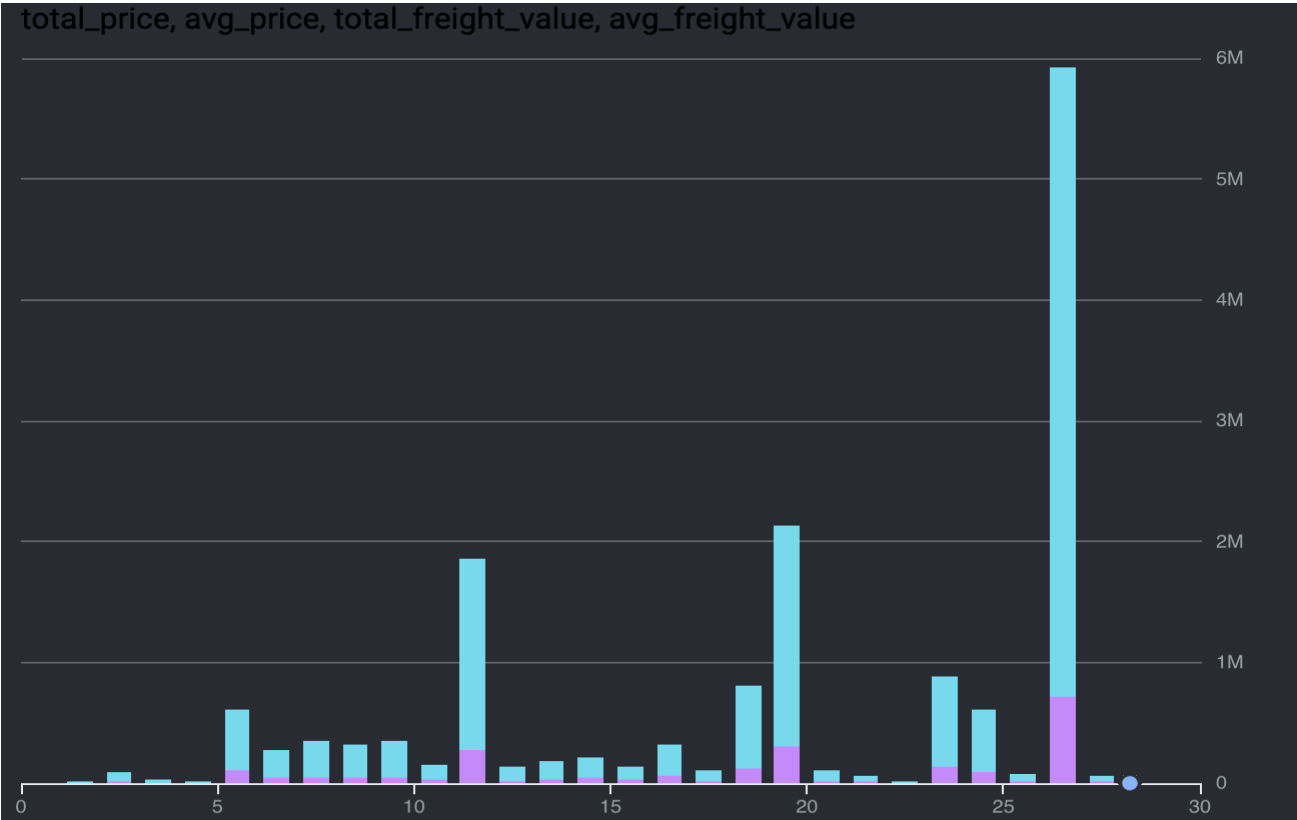
SQL Query

```
SELECT
c.customer_state,
ROUND(SUM(o2.price), 0) AS total_price,
ROUND(AVG(o2.price), 0) AS avg_price,
ROUND(SUM(o2.freight_value), 0) AS total_freight_value,
ROUND(AVG(o2.freight_value), 0) AS avg_freight_value
FROM `target_sql.orders` AS o1
JOIN `target_sql.order_items` AS o2
ON o1.order_id = o2.order_id
JOIN `target_sql.customers` AS c
ON o1.customer_id = c.customer_id
GROUP BY c.customer_state
ORDER BY c.customer_state;
```

Query Output

Row	customer_state	total_price	avg_price	total_freight_value	avg_freight_value
1	AC	15983.0	174.0	3687.0	40.0
2	AL	80315.0	181.0	15915.0	36.0
3	AM	22357.0	135.0	5479.0	33.0
4	AP	13474.0	164.0	2788.0	34.0
5	BA	511350.0	135.0	100157.0	26.0
6	CE	227255.0	154.0	48352.0	33.0
7	DF	302604.0	126.0	50625.0	21.0
8	ES	275037.0	122.0	49765.0	22.0
9	GO	294592.0	126.0	53115.0	23.0
10	MA	119648.0	145.0	31524.0	38.0

Visualization



Business Insight

Significant variation exists between states in both average order values and freight expenses, reflecting differences in customer purchasing power and logistics complexity.

Recommendation

Adopt region-specific pricing, shipping, and promotional strategies.

Potential Business Impact

Optimized regional strategies can improve margins while maintaining competitiveness.

Question 11

How long does it take to deliver each order, and how does that compare to the estimated delivery date?

Business Objective

This measures actual delivery performance against customer expectations, which is one of the strongest drivers of satisfaction and repeat purchase in e-commerce.

SQL Query

```
SELECT
order_id,
DATE_DIFF(DATE(order_delivered_customer_date),DATE(order_purchase_timestamp),DAY)
AS time_to_deliver,
DATE_DIFF(DATE(order_delivered_customer_date),DATE(order_estimated_delivery_date),DAY)
AS diff_estimated_delivery
FROM `target_sql.orders`
WHERE order_status='delivered';
```

Query Output

Row	order_id	time_to_deliver	diff_estimated_d...
1	bfbd0f9bdef84302105ad712db...	55	36
2	98974b076b01553d49ee64679...	44	-7
3	c4b41c36dd589e901f6879f25a...	36	-15
4	d2292ff2201e74c5db154d1b7a...	30	-21
5	95e01270fcbae986342340010...	31	-20
6	ed8c7b1b3eb256c70ce0c7423...	45	-6
7	5cc475c7c03290048eb2e742c...	69	18
8	6b3ee7697a02619a0ace2b3f0a...	48	-3
9	3b2ca3293a7ce539ea2379d70...	43	-8
10	b2f92b2f7047cd8b35580d629d...	43	-8

Business Insight

Most delivered orders arrive within a reasonable window of their purchase date, and the majority arrive before the estimated delivery date, which suggests the estimation model is somewhat conservative. However, a tail of orders shows significantly longer delivery times, which likely drives a disproportionate share of negative reviews.

Recommendation

Investigate the specific orders and regions that fall in the long delivery tail, and consider tightening estimated delivery dates in regions that consistently deliver early, which can be used as a competitive advantage in marketing.

Potential Business Impact

Reducing the delivery time tail directly reduces the volume of dissatisfied customers, while more accurate estimates in fast delivering regions can be used to boost conversion by promising and beating faster delivery windows.

Question 12

Which states have the highest and lowest average freight value?

Business Objective

This narrows the state level freight analysis down to a clear top five and bottom five list, making it easy to prioritize action.

SQL Query

Highest average freight value

```
SELECT c.customer_state,
ROUND(AVG(freight_value),2) AS avg_freight_value
FROM `target_sql.orders` AS o1
JOIN `target_sql.order_items` AS o2
ON o1.order_id = o2.order_id
JOIN `target_sql.customers` AS c
ON o1.customer_id = c.customer_id
GROUP BY c.customer_state
ORDER BY avg_freight_value DESC
LIMIT 5;
```

Lowest average freight value

```
SELECT c.customer_state,
ROUND(AVG(freight_value),2) AS avg_freight_value
FROM `target_sql.orders` AS o1
JOIN `target_sql.order_items` AS o2
```

```

ON o1.order_id = o2.order_id
JOIN `target_sql.customers` AS c
ON o1.customer_id = c.customer_id
GROUP BY c.customer_state
ORDER BY avg_freight_value ASC
LIMIT 5;

```

Query Output

Row	customer_state	avg_freight_value
1	RR	42.98
2	PB	42.72
3	RO	41.07
4	AC	40.07
5	PI	39.15

Highest

Row	customer_state	avg_freight_value
1	SP	15.15
2	PR	20.53
3	MG	20.63
4	RJ	20.96
5	DF	21.04

Lowest

Business Insight

Remote states experience substantially higher freight costs compared to states located near major distribution centers.

Recommendation

For the five highest freight states, explore regional courier partnerships or a satellite warehouse, and consider whether free shipping thresholds need to be adjusted specifically for these regions to protect margin.

Potential Business Impact

Targeted freight cost reduction in just five states can meaningfully improve overall logistics margin without requiring a nationwide infrastructure overhaul.

Question 13

Which states have the highest and lowest average delivery time?

Business Objective

This identifies the specific states where delivery speed is a strength or a weakness, supporting focused operational improvements.

SQL Query

Highest average delivery time

```
SELECT c.customer_state,
AVG(EXTRACT(DATE FROM o1.order_delivered_customer_date) - EXTRACT( DATE FROM
o1.order_purchase_timestamp)) AS avg_time_to_delivery
FROM `target_sql.orders` AS o1
JOIN `target_sql.order_items` AS o2
ON o1.order_id = o2.order_id
JOIN `target_sql.customers` AS c
ON o1.customer_id = c.customer_id
GROUP BY c.customer_state
ORDER BY avg_time_to_delivery DESC
LIMIT 5;
```

Lowest average delivery time

```
SELECT c.customer_state,
AVG(EXTRACT(DATE FROM o1.order_delivered_customer_date) - EXTRACT( DATE FROM
o1.order_purchase_timestamp)) AS avg_time_to_delivery
FROM `target_sql.orders` AS o1
JOIN `target_sql.order_items` AS o2
ON o1.order_id = o2.order_id
JOIN `target_sql.customers` AS c
ON o1.customer_id = c.customer_id
GROUP BY c.customer_state
ORDER BY avg_time_to_delivery ASC
LIMIT 5;
```

Query Output

Row	customer_state	avg_time_to_delivery
1	AP	0-0 28 5:20:0
2	RR	0-0 28 4:10:26.086956
3	AM	0-0 26 8:5:53.374233
4	AL	0-0 24 10:44:7.306791
5	PA	0-0 23 16:51:0.341555

Row	customer_state	avg_time_to_delivery
1	SP	0-0 8 15:53:38.629287
2	PR	0-0 11 21:26:1.975570
3	MG	0-0 11 22:5:50.607726
4	DF	0-0 12 21:27:8.025477
5	SC	0-0 14 22:48:18.975109

Business Insight

States with higher freight costs generally also experience longer delivery times, indicating logistical inefficiencies in remote areas.

Recommendation

Treat the slowest delivery states as a single priority group for both freight and delivery time improvements, since a shared root cause means a shared solution, such as a regional hub, is likely to fix both problems at once.

Potential Business Impact

Improving delivery time in the slowest states can reduce delivery related complaints and improve review scores in exactly the regions where customer experience currently lags the rest of the country.

Question 14

Which states have the fastest deliveries compared to their estimated delivery date?

Business Objective

This highlights where the business is most reliably beating customer expectations, which is valuable both operationally and as a marketing proof point.

SQL Query

```
SELECT
c.customer_state,
ROUND(AVG(DATE_DIFF(DATE(o.order_estimated_delivery_date),DATE(o.order_delivered_customer_date),DAY)),2)
  AS avg_days_early
FROM `target_sql.orders` o
JOIN `target_sql.customers` c
ON o.customer_id = c.customer_id
WHERE o.order_status='delivered'
GROUP BY c.customer_state
ORDER BY avg_days_early DESC
LIMIT 5;
```

Query Output

Row	customer_state	avg_days_early
1	AC	20.72
2	RO	20.1
3	AP	19.69
4	AM	19.57
5	RR	17.29

Business Insight

Several states consistently receive orders earlier than estimated delivery dates, indicating conservative delivery commitments.

Recommendation

Use these states as a benchmark to understand what is working well operationally, and consider whether the estimated delivery date shown to customers in these states can be tightened to create a more competitive, faster looking promise at checkout.

Potential Business Impact

A tighter, more accurate delivery promise in these states can be used as a conversion driver at checkout, since faster estimated delivery is a well known purchase decision factor for online shoppers.

Question 15

How have payment type preferences trended over time?

Business Objective

This tracks how customers pay for their orders month by month, which supports decisions on payment partnerships, checkout design, and promotional financing offers.

SQL Query

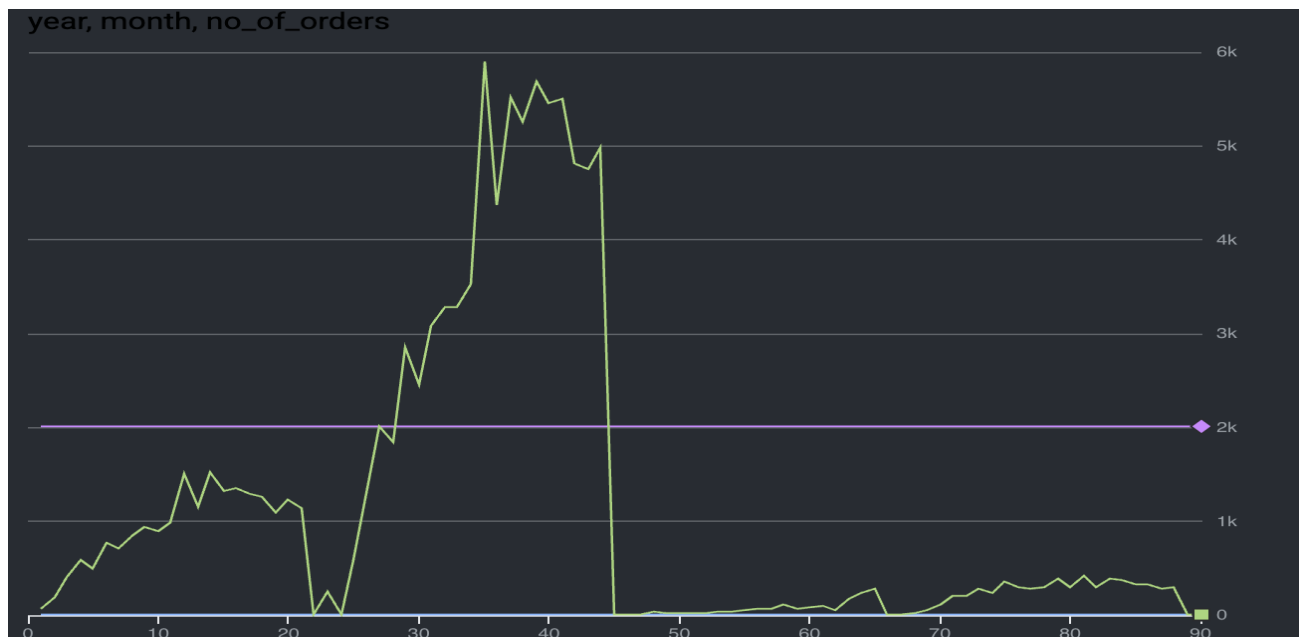
```
SELECT
p.payment_type,
EXTRACT(YEAR FROM o.order_purchase_timestamp) AS year,
EXTRACT(month FROM o.order_purchase_timestamp) AS month,
COUNT(o.order_id) AS no_of_orders
```

```
FROM `target_sql.orders` AS o
INNER JOIN `target_sql.payments` AS p
ON o.order_id = p.order_id
GROUP BY p.payment_type, year, month
ORDER BY p.payment_type, year, month;
```

Query Output

Row	payment_type	year	month	no_of_orders
1	UPI	2016	10	63
2	UPI	2017	1	197
3	UPI	2017	2	398
4	UPI	2017	3	590
5	UPI	2017	4	496
6	UPI	2017	5	772
7	UPI	2017	6	707
8	UPI	2017	7	845
9	UPI	2017	8	938
10	UPI	2017	9	903

Visualization



Business Insight

Credit cards dominated transactions throughout the analysis period, while digital and alternative payment methods also showed steady usage growth, indicating increasing customer preference for convenient cashless payments.

Recommendation

Target should continue optimizing the credit card checkout experience while maintaining support for alternative digital payment methods to cater to evolving customer preferences.

Potential Business Impact

Maintaining strong support for both major payment methods protects conversion rates across the full range of the customer base rather than only the credit card using segment.

Question 16

How are orders distributed based on the number of payment installments used?

Business Objective

This shows how customers prefer to split their payments, which is important in Brazil where installment based buying is common, and informs decisions about minimum order values for installment offers.

SQL Query

```
SELECT
payment_installments,
COUNT(order_id) AS no_of_orders
FROM `target_sql.payments`
GROUP BY payment_installments
ORDER BY payment_installments
```

Query Output

Row	payment_installm...	no_of_orders ▼
1	0	2
2	1	52546
3	2	12413
4	3	10461
5	4	7098
6	5	5239
7	6	3920
8	7	1626
9	8	4268
10	9	644

Business Insight

A large share of orders are paid in a single installment, but a meaningful portion of customers choose to split payments across multiple installments, particularly for higher value orders, which reflects the popularity of installment based buying in the Brazilian market.

Recommendation

Promote installment options more visibly at checkout for higher value carts, since customers who are offered a comfortable installment plan are more likely to complete a higher value purchase rather than abandoning the cart.

Potential Business Impact

Better visibility of installment options can increase average order value and reduce cart abandonment, particularly for higher priced product categories.

6. Overall Findings

The analysis of Target Brazil's e-commerce operations reveals a business that experienced strong expansion during the observed period while simultaneously facing several operational and logistical challenges associated with scaling a nationwide marketplace.

The order trend analysis indicates a clear increase in customer demand over time, suggesting growing acceptance of online shopping and successful expansion of Target's digital presence in Brazil. The existence of monthly fluctuations in order volume also highlights the presence of seasonality in customer purchasing behavior, making demand forecasting and inventory planning increasingly important for maintaining service quality.

Geographical analysis demonstrates that customer demand is not evenly distributed across Brazil. Certain states contribute significantly more customers and orders than others, reflecting regional differences in population density, purchasing power, economic activity, and internet penetration. This uneven distribution suggests that a uniform national strategy may not be the most effective approach for growth and resource allocation.

The logistics analysis highlights one of the most important operational findings of the study. Freight costs and delivery times vary considerably across states, indicating the influence of geographical distance and transportation infrastructure on overall business performance. States with higher freight costs often also experience longer delivery times, suggesting that logistics efficiency remains one of the key factors influencing customer experience and profitability.

Delivery performance analysis also shows that in several regions orders are consistently delivered earlier than the estimated delivery date. While this reflects strong operational execution, it may also indicate that delivery estimates are overly conservative and could potentially be optimized to improve customer expectations and conversion rates.

Payment behavior analysis reveals a clear customer preference for certain payment methods and installment options. The popularity of installment payments suggests that financial flexibility plays an important role in purchase decisions within the Brazilian e-commerce market. This creates opportunities for improving conversion rates and increasing average order values through better financing options.

Overall, the analysis demonstrates that Target Brazil has successfully established a broad national presence and achieved strong growth, but future expansion will depend heavily on improvements in logistics efficiency, regional strategy development, and customer experience optimization.

7. Strategic Recommendations

Based on the findings of this analysis, several strategic initiatives can be recommended to improve profitability, operational efficiency, and customer satisfaction.

The first priority should be logistics optimization in regions with high freight costs and long delivery times. Since freight expenses directly impact profitability and customer satisfaction, Target should explore options such as regional fulfillment centers, local distribution hubs, or partnerships with third-party logistics providers in underperforming regions. Even moderate reductions in shipping costs could significantly improve margins at scale.

A second recommendation involves adopting a regional rather than national operating strategy. Customer demand varies considerably across states, and therefore marketing investments, inventory allocation, and expansion efforts should reflect local market conditions rather than applying identical approaches across all regions.

Demand forecasting capabilities should also be strengthened by leveraging the seasonality patterns identified in the analysis. Anticipating demand spikes allows the business to prepare inventory, warehouse staffing, and delivery capacity in advance, reducing the risk of stockouts and delayed deliveries during peak periods.

The findings related to delivery estimation suggest an opportunity to improve customer communication and trust. In states where orders consistently arrive earlier than estimated, Target should consider reducing estimated delivery windows to provide customers with more accurate expectations. Delivery speed remains one of the most important competitive factors in e-commerce, and improved delivery promises can positively influence purchasing decisions.

Payment strategy optimization represents another important opportunity. Since customers demonstrate a strong preference for specific payment methods and installment plans, these options should be prominently displayed throughout the purchasing process. Additional installment options for higher-value purchases may increase average order value and reduce cart abandonment.

Marketing activity should also be aligned with customer purchasing behavior identified through the time-of-day analysis. Promotions, email campaigns, and notifications delivered during periods of high customer activity are likely to achieve better engagement and conversion rates compared to randomly timed campaigns.

Finally, Target should continue investing in customer data collection and analytics capabilities. As the business grows, data-driven decision making will become increasingly important for managing costs, improving customer experience, and identifying new growth opportunities.

8. Conclusion

This project successfully applied SQL and BigQuery to analyze customer behavior, order trends, payment patterns, logistics performance, and delivery efficiency within Target Brazil's e-commerce ecosystem.

The analysis demonstrates that while the business has achieved strong growth and nationwide reach, expansion at scale introduces new operational complexities that require data-driven decision making. Geographic variation in customer demand, freight costs, and delivery performance highlights the importance of localized business strategies and targeted investments.

The findings also emphasize the critical role of logistics in e-commerce success. Delivery speed, freight expenses, and estimation accuracy directly influence customer satisfaction and long-term retention. Improving these areas can generate significant competitive advantages in an increasingly crowded online marketplace.

Payment behavior analysis further reveals that understanding customer preferences is essential for maximizing conversion rates and increasing average order values. Flexible payment options and installment plans are not simply convenience features but core components of the customer purchasing journey in the Brazilian market.

Perhaps the most important outcome of this project is the demonstration of how transactional data can be transformed into actionable business insights. SQL analysis alone does not create business value; value is generated when data is used to support operational improvements, strategic decisions, and customer-focused initiatives.

The project highlights the importance of combining technical analytical skills with business understanding to solve real-world problems. The insights and recommendations generated through this analysis provide a strong foundation for future decision making related to logistics optimization, customer acquisition, payment strategy, and operational scalability.

As Target Brazil continues to expand, maintaining a strong analytical culture and leveraging data-driven decision making will be essential for sustaining growth, improving profitability, and delivering superior customer experiences across the country.